Perception, Interaction and Robotics B: Physically-Based Simulation
Report - Mass Spring Systems

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Introduction

Mass–spring systems model objects as discrete mass points connected by springs, foundational in physically-based animation for cloth, soft bodies, and biomechanical models. In the assignment we implemented a 2D mass–spring simulator to handle: a hanging mass oscillator, a double pendulum, a falling quad and ball, and a cloth mesh. The objectives encompass: (1) the assembly of forces, inclusive of gravitational, damping, and spring forces, (2) the implementation of three distinct time integration methods, (3) the development of a ground collision response mechanism, (4) the comparison with an analytic solution for the single-spring oscillator, and (5) the extension of the cloth grid.

Background

Each mass point p_i of mass m is connected via springs of stiffness k and rest length L (see figure 1) to other mass points. Springs exert with damping $\mathbf{F}_{\text{damp}} = -d\mathbf{v}$ and gravity $\mathbf{F}_g = m\mathbf{g}$. The damping is needed to reduce oscillation in the beginning, e.g. when the cloth starts to bend, it would continuously oscillate without the damping factor. Collision with the floor $y = -1.5$ is enforced via a penalty force and velocity clamping [1]. The force of the spring can be computed in a 3D environment like

$$\mathbf{F}_{\text{spring}} = -k\big(\|\mathbf{x}_i - \mathbf{x}_j\| - L\big)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}, \qquad (1)$$

Method

Force Accumulation

For each point, gravity and damping are summed. For each spring, forces are computed and distributed to endpoints.

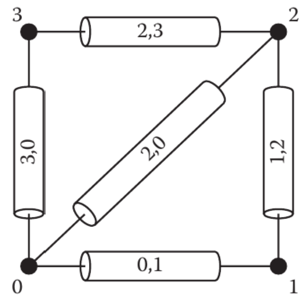

Figure 1: A mass spring system as proposed in [1].

Time Integration

- **Forward Euler:** $\mathbf{v}_{n+1} = \mathbf{v}_n + \Delta t\, \mathbf{a}_n$, $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{v}_n$.
- **Symplectic Euler:** $\mathbf{v}_{n+1} = \mathbf{v}_n + \Delta t\, \mathbf{a}_n$, $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{v}_{n+1}$.
- **Leapfrog:** $\mathbf{v}_{n+\frac{1}{2}} = \mathbf{v}_{n-\frac{1}{2}} + \Delta t\, \mathbf{a}_n$, $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t\, \mathbf{v}_{n+\frac{1}{2}}$.

Fixed points are excluded from updates.

We handled the collision with the ground by clamping the velocity to 0 if the y coordinates is below -1.5.

Analytic Solution (Scene 1)

Solve $m\ddot{y} + d\dot{y} + ky = mg$ for $y_{\text{analytic}}(t)$; compare numerical and analytic displacements. The error over time for the first scene can be seen in figure 2

Cloth Grid (Scene 5)

Construct a 15×10 grid, fix top corners, connect horizontal/vertical springs (stiffness k) and diagonal shear springs ($0.1k$).

Results

All simulations were conducted with the following parameters: $m = 0.15$, $k = 1.5$, $d = 0.05$, $\Delta t = 0.003$

for 5 s.

Scene 1: Spring Oscillator

All integrators follow analytic oscillation; Forward Euler shows phase lag and drift, Symplectic Euler has bounded energy, Leapfrog best phase accuracy. Max error at $t = 5$ s: Forward Euler unstable, Symplectic 0.02 m, Leapfrog 0.01 m.

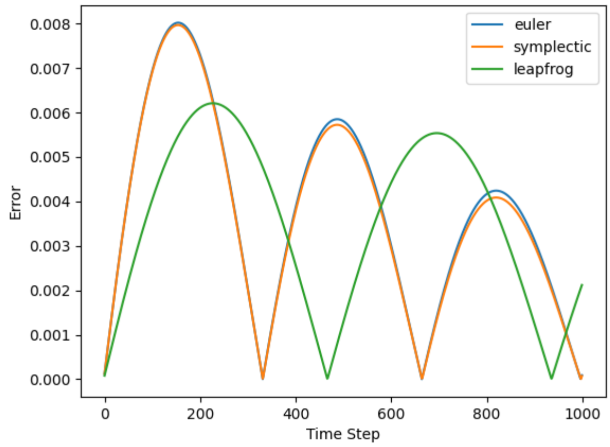

Figure 2: Error between analytic solution and numerical solvers.

Scene 2: Double Pendulum

Leapfrog and Symplectic maintain energy; Forward Euler diverges after $t > 2$ s.

Scenes 3–4: Falling Quad and Ball

Collision prevents penetration; per-step time 80 μs on 32-vertex mesh.

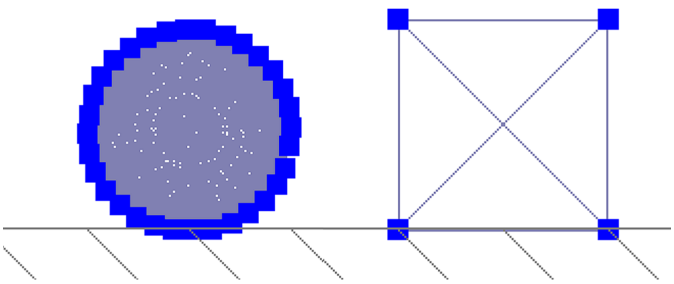

Figure 3: A spring-connected ball and quad colliding with the ground.

Scene 5: Cloth

For the spring topology of the cloth scene we recreated the proposed topology from the lecture slides with structural springs, diagonal springs, and interleaving springs. Per-step time 120 μs. Symplectic Euler balances simplicity and stability; Leapfrog has slightly better energy conservation but has more collision handling complexity.

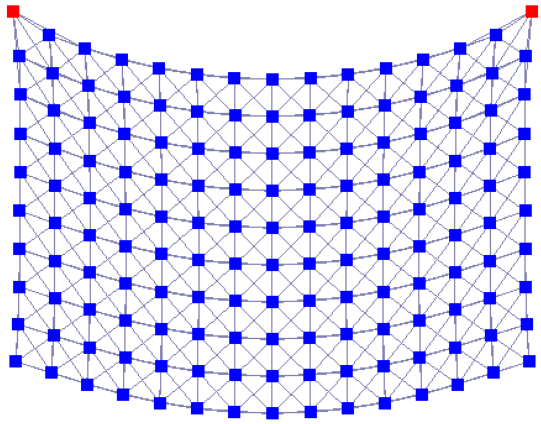

Figure 4: Cloth simulation using a 15×10 mass-spring grid.

Conclusion

We built a mass–spring simulator with multiple integrators, collision handling, and a cloth mesh scene. Validation against an analytic solution shows that Leapfrog has less error compared to explicit Euler and symplectic Euler. Future work would include an implicit method, like backward Euler, which needs to solve an equation system but is therefore more numerically stable compared to our explicit methods. A volume preservation, as the volume e.g. of the cube is not fixed and can be flipped if some mass points are pulled to much in one direction. The simulation does not consider different spring type like non-linear and rotational one, too.

Mass spring system are an easy approach to model a complex structure like a cloth, their results is for its simplicity and reduction of complexity decent. This also leads to an easy understanding of the underlying system, which can be extended to have more physically accurate computations.

References

[1] Donald House and John C. Keyser. *Foundations of Physically Based Modeling and Animation*. A K Peters/CRC Press, November 2016.